

Spectral weight control in physics-informed neural networks: what the orthogonality penalty does, and how it transfers to shock-dominated shallow-water flow

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Abstract

Wang et al. [8] improve physics-informed neural network accuracy by factorising each hidden weight at initialisation, training the left factor and the singular values in place of the weight, and penalising departures of the left factor from orthogonality with a hand-tuned weight w_U . Their account assigns the gain to control over the weight spectrum. We reproduce their advection case over 20 seeds and then measure the trained factors, which their paper does not report. At the published $w_U = 1/35000$ the trained left factor is not close to orthogonal — $\|U^\top U - I\|_2$ has median 0.98, indicating a collapsed eigendirection — and the trained singular values differ from the effective weight’s actual singular values by up to a factor of 13. Constraining the factor to be exactly orthogonal, which removes w_U entirely, reaches only 1.7×10^{-2} median relative error against the penalty’s 8.4×10^{-3} ; removing the penalty from the same parameterisation returns the error to 2.1×10^{-1} , worse than an unconstrained network’s 8.4×10^{-2} . Reading the term as a bound on the factor’s scale predicts that a weaker bound should also work; we test that by pinning only $\sigma_{\max}(U)$, and it is the worst regime of the six at 3.7×10^{-1} median, so what the penalty supplies is a pull of the whole spectrum toward 1 rather than a bound on scale. Transferring the comparison to two-dimensional shallow-water dam breaks, where the solution carries a shock and the bounded-gradient hypothesis of the source paper’s Theorem 2.1 fails, no spectral regime separates from an unmodified network by more than about 10% in L^1 depth error on the three shock cases, and w_U can be moved three decades without measurable effect. On the one smooth initial condition in the set, exact orthogonality and spectrum-only training are a factor of 2.5 worse than an unmodified network. Establishing this required checking our finite-volume reference against an independently implemented solver family, which disagrees with it by more than the differences under test on two of three benchmarks — a floor that comparisons of this kind do not usually report.

1 Introduction

Physics-informed neural networks solve a PDE by minimising its residual at collocation points [5]. They fail on targets with steep or high-frequency structure, and a line of work attributes the failure to properties of the network’s Jacobian. Wang et al. [8] formalise this through a quantity they call the condition number, $\kappa(f) = \|J(f)\|_F$, and propose a remedy: decompose each hidden weight

$W_j = U_j S_j V_j^\top$ once at initialisation, freeze V_j , train U_j and $\text{diag}(S_j)$ in place of W_j , and add

$$w_U \sum_j D_j^2, \quad D_j = \|U_j^\top U_j - I\|_2 \quad (1)$$

to the loss so that U_j stays orthogonal and $\text{diag}(S_j)$ therefore carries the weight’s spectrum.

Their conclusion lists three open problems. The penalty weight w_U is tuned per equation across two and a half decades, from 1/600 to 1/200000 over five PDEs, and their Section 4.6.3 reports that NTK- and DB-PINN-style adaptive weighting diverges on this term because the penalty starts at exactly zero. The penalty costs 1.28 to 1.83 times vanilla training time across their width and collocation sweeps. All five validation problems are, in their own description, controlled synthetic examples, and all five are smooth.

That last point has a specific consequence. Every hypothesis of their Theorem 2.1 requires $u \in H^1$ with $\|J(u)\|_F$ bounded. A dam-break solution carries a shock, so $\|J(u)\|_F$ is unbounded and $u \notin H^1$: the theorem is formally inapplicable to the entire problem class. The authors say as much in a different context, noting that the condition is violated once training collapses.

We take the three limitations in order and add the excluded regime. Section 5 reproduces the advection case and then measures what the penalty does to the trained factors, which decides what kind of constraint the term is. Section 6 moves the same weight regimes to two-dimensional shallow-water dam breaks. Section 6.5 reports a result we did not set out to find: on two of three benchmarks the reference against which all of this is measured is less certain than the differences being measured.

2 The parameterisation and the regimes compared

Write a hidden layer’s weight as $W \in \mathbb{R}^{m \times n}$ and its initial factorisation as $W_0 = U_0 S_0 V_0^\top$. All regimes below keep $V = V_0$ frozen and differ in how U and the spectrum are treated. A fifth, introduced in Section 5.3 once the measurements motivate it, bounds U ’s largest singular value.

Unconstrained. W trained directly. The baseline.

Penalised spectral. U and $s = \text{diag}(S)$ trained, with Eq. (1) added to the loss. This is the published method.

Exactly orthogonal. U constrained to the Stiefel manifold by the matrix-exponential parameterisation of Lezcano-Casado and Martínez-Rubio [3], and s trained. The penalty and w_U do not exist in this regime, which is what makes it the direct test of the first stated limitation.

Spectrum only. U and V both frozen at their orthogonal initial values, s alone trained. Structurally this layer is the one used by SVD-PINNs [2]; there it transfers a trained network to a new equation, whereas here it is trained from scratch. The construction is not ours and is not presented as new.

The distinction between the second and third regimes is the paper’s pivot. If the penalty is an approximation to orthogonality, then enforcing orthogonality exactly should do at least as well and costs no tuned weight. If it is something else, the two will separate.

3 Related work

Spectral structure in network weights. Spectral normalisation [4] divides a weight by its largest singular value to force a 1-Lipschitz map. It acts on the same object as the method studied here and with the opposite intent: it caps the gradient magnitude a network can express, whereas the diagnostic motivating [8] holds that networks fail on steep targets by expressing too little of it. The exactly-orthogonal regime we test uses the matrix-exponential parametrisation of Lezcano-Casado and Martínez-Rubio [3], introduced for recurrent networks; we are not aware of a prior application to PINN training, and the result in Section 5 suggests why one might not have been reported. SVD-PINNs [2] freeze both orthogonal factors and retrain the singular values to move a trained network to a new equation. Our spectrum-only regime is that layer trained from scratch rather than transferred; the construction is theirs, and what we add is its failure mode as a from-scratch method.

Constraint handling in PINN losses. AL-PINNs [6] apply an augmented-Lagrangian treatment to the initial and boundary conditions — constraints on the solution, not on weight structure. The distinction matters here because the adaptive-weighting schemes that [8] report as diverging on their penalty are of this family, and the reason they diverge is specific to a penalty that starts at exactly zero.

PINNs at shocks. A separate line accepts that pointwise residuals are ill-defined at a discontinuity and changes the objective or the architecture: weak- and entropy-form losses, jump conditions imposed through the architecture, gradient-weighted losses concentrated near shocks, and relaxation formulations. None of these measures or controls a property of the network’s weights, so the comparison in Section 6 is orthogonal to them rather than competing with them; combining a spectral parametrisation with a weak-form loss is untested and outside what we report here. The shallow-water PINN literature specific to dam breaks is surveyed in [9], from which our benchmarks and finite-volume harness are taken.

What we did not find. A search in August 2026 returned no PINN work training weights under an exact orthogonality constraint, no condition-number or gradient-norm mismatch diagnostic for PINNs on hyperbolic problems, and no indexed citation of [8], which appeared online two months earlier. The searches were run against arXiv-indexed material and are not exhaustive; the sweep needs repeating before submission.

4 Implementation

The penalty in Eq. (1) uses the matrix 2-norm, following the source paper. An early version of our code used the Frobenius norm, which at width 128 is up to 128 times stronger. Both are reported: the dam-break matrix of Table 3 was produced with the Frobenius penalty and the sweep of Table 4 with Eq. (1), and Section 6.3 compares them directly. The choice turns out to leave integrated depth error alone and to change where the shock is placed. All results here use L-BFGS at 3000 iterations for the advection case and Adam at 10^{-3} for 20000 iterations on the shallow-water cases. Seeds are set before model construction, so initial weights are reproducible.

5 Reproduction and mechanism on the advection case

5.1 The gate

Table 1 repeats the source paper’s Section 4.1 advection setup: a 3×25 tanh network, Xavier initialisation, 100 boundary and 500 interior points by Latin hypercube sampling, L-BFGS to 3000 iterations, $w_U = 1/35000$. We report 20 seeds. Distributions in this problem are heavy-tailed — the unconstrained network’s worst seed is 53 times its best — so the median is the more stable summary and both are given.

Table 1: Relative L^2 error on the advection case over 20 seeds. The first two rows are the comparison the source paper reports (their Table 5, tanh activation); the last four are regimes it does not test. Reproducing the published cells is what licenses reading the rest.

Regime	Mean $\pm$ s.d.	Median	Wang et al. Table 5
Unconstrained	$(1.37 \pm 1.4) \times 10^{-1}$	8.4×10^{-2}	$(1.13 \pm 1.67) \times 10^{-1}$
Penalised, $w_U = 1/35000$	$(1.46 \pm 1.5) \times 10^{-2}$	8.4×10^{-3}	$(5.64 \pm 3.86) \times 10^{-3}$
Exactly orthogonal	$(3.31 \pm 3.9) \times 10^{-2}$	1.7×10^{-2}	—
Spectrum only	$(3.29 \pm 5.5) \times 10^{-1}$	1.2×10^{-1}	—
Bounded $\sigma_{\max}(U)$	$(4.68 \pm 5.1) \times 10^{-1}$	3.7×10^{-1}	—
Same parameterisation, $w_U = 0$	$(3.37 \pm 3.5) \times 10^{-1}$	2.1×10^{-1}	—

The two published cells reproduce within their reported spread, so the setup is faithful. Three readings follow from the rest. Exact orthogonality, which eliminates w_U , costs about a factor of two in median error against the tuned penalty. Training the spectrum alone is worse than an unconstrained network and diverges on 2 of 20 seeds, so the trained directions rather than the trained spectrum carry the gain — a negative result for the cheap variant, and one that already sits awkwardly with a spectral explanation of the mechanism. Removing the penalty while keeping the parameterisation is the worst regime bar the bounded one of Section 5.3.

5.2 What the penalty does to the trained factors

The source paper reports errors, not factor geometry. We measure, per factorised layer of trained networks over 3 seeds, the orthogonality defect $D = \|U^\top U - I\|_2$ and the largest relative gap between the trained $|s|$ and the singular values of the effective weight $USV^\top$ (Table 2).

Table 2: Factor geometry after training, 3 seeds, median over layers. If the described mechanism held, D would be near zero and the two spectra would coincide. The bounded regime is discussed in Section 5.3; note that it matches the penalised regime’s D while missing the spectrum by two further orders of magnitude.

Setting	Median error	Median D	$ s $ vs. true singular values
$w_U = 1/35000$	7.9×10^{-3}	0.98	up to $13\times$ apart
$w_U = 0$	3.05×10^{-1}	63 (max 177)	up to $31\times$ apart
Bounded $\sigma_{\max}(U)$	3.8×10^{-1}	1.00	up to $1290\times$ apart

At the published weight $D \approx 1$, which for a defect measured in the 2-norm means an eigendirection of $U^\top U$ has collapsed: the trained factor is rank-deficient, not near-orthogonal. The trained

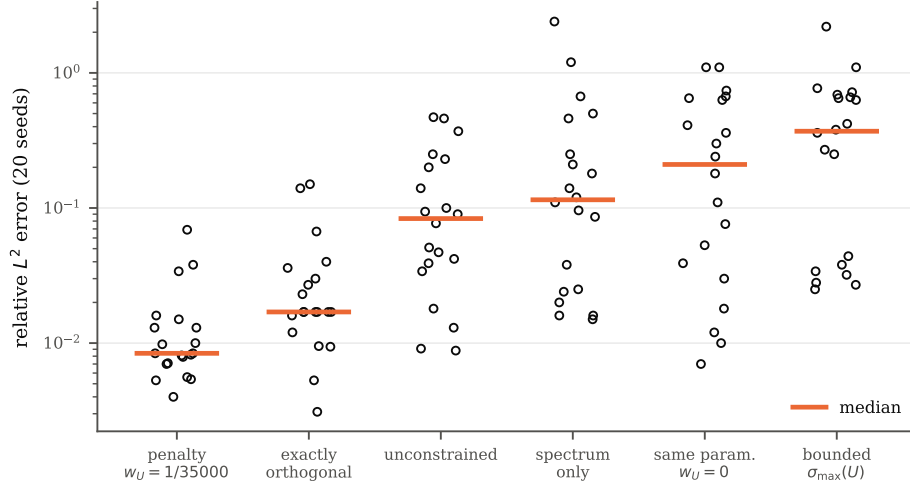


Figure 1: Per-seed relative error on the advection case, 20 seeds per regime, log scale, ordered by median. The distributions are heavy-tailed and overlap heavily, which is why medians are quoted alongside means. Bounding $\sigma_{\max}(U)$ — the weaker constraint predicted to substitute for the penalty — is the worst regime, below even removing the penalty outright.

s is not the effective weight’s spectrum. Both premises of the stated mechanism fail, while the method’s accuracy advantage holds.

Ordering the regimes by median error gives unconstrained-parameterised 2.1×10^{-1} , spectrum-only 1.2×10^{-1} , unconstrained 8.4×10^{-2} , exactly orthogonal 1.7×10^{-2} , weakly penalised 8.4×10^{-3} . The penalty is doing something the method needs: removing it lets the defect grow by two orders of magnitude and destroys the gain, while enforcing the constraint it nominally approximates recovers only about half of it. Exact orthogonality is a stronger constraint than the problem wants. The natural reading is that the term bounds the scale of U rather than orthogonalising it, and Section 5.3 tests that reading directly — it does not survive.

This reframes the first stated limitation: the open question is not how to adapt w_U but which constraint class the term belongs to. The ordering above suggested that a bound weaker than orthogonality — a cap on U ’s largest singular value, or a box on all of them — should recover the gain without a tuned penalty. We implemented the first of those, and it does not.

5.3 Testing the prediction: bounding the largest singular value

The bounded regime rescales U on every forward pass so that $\sigma_{\max}(U) = 1$, leaving the remaining singular values free in $[0, 1]$. It is strictly weaker than orthogonality, which fixes all of them at 1, and it admits the rank-deficient factor the penalty was measured to produce. It carries no penalty term and no w_U . The scale it removes is not lost to the layer, since $W = U \text{diag}(s) V^\top$ and s is trained.

Over 20 seeds it reaches $(4.68 \pm 5.1) \times 10^{-1}$ mean and 3.7×10^{-1} median relative error: the worst of the six regimes, below even the unpenalised parameterisation. The prediction is refuted.

Measuring the bounded networks’ factors explains the failure and corrects our earlier reading (last row of Table 2). The bounded regime reaches $D = 1.00$, essentially the same defect as the penalised regime’s 0.98 — so on the statistic we used to characterise the penalty, the two are indistinguishable. They are not indistinguishable in behaviour: the gap between the trained $|s|$ and

the effective weight’s singular values is $13\times$ under the penalty and $1290\times$ under the cap, and the bounded networks drive their smallest trained singular value to 0 exactly.

The reason is that $D = \max_i |\sigma_i(U)^2 - 1|$ saturates as soon as a single direction of U collapses, so it cannot distinguish one collapsed direction from many. Capping $\sigma_{\max}(U)$ reproduces the value of D while leaving the rest of the spectrum free to collapse, and the effective weight then has nothing to do with the spectrum s nominally controls. The conclusion is narrower than the one we drew from the accuracy ordering alone: what the penalty supplies is not a bound on the factor’s scale — a bound on scale is exactly what the cap provides, and it fails — and D is too coarse a summary to say what it is instead. Section 5’s reading was underdetermined by its own evidence.

The other half of the prediction — a box holding every singular value of U in $[1 - \varepsilon, 1 + \varepsilon]$ — follows from that reading and is untested. It is the constraint class that matches the measured behaviour, with ε a geometrically interpretable knob in place of a loss weight on an unnormalised penalty, and it is the experiment we would run next.

The evidence here is one smooth one-dimensional problem, one architecture and one optimiser. Whether the ordering survives on a shock is a separate question, which is what Section 6 asks.

6 Shallow-water dam breaks

6.1 Setup

The benchmarks are two-dimensional dam breaks over a Gaussian-hump bed at the conventions of the comparative study of Zahra et al. [9]: domain $[0, 100]^2$ m, $g = 2$ m/s², reflective walls on all four sides, $t_{\text{end}} = 2$ s. We use a step release, a circular column over a wet bed, the same column over a dry bed, and a smooth Gaussian mound, all with Manning $n = 0$. A fifth case, three submerged humps with Manning friction over 20 s, is included and then set aside for the reason given below. Errors are L^1 and L^2 of depth at the final output time on a shared 128^2 evaluation grid, against a reference computed in-process with an HLLC MUSCL–van Leer finite-volume scheme [7] at 512^2 with hydrostatic reconstruction for well-balancing [1]. Baselines are an unmodified network, a Fourier-feature network — the standard remedy for high-frequency targets, absent from the source paper’s comparison — and a finite-volume-informed network. Five seeds per cell.

Every strong-form entry is a 6-layer, 128-unit tanh network mapping (x, y, t) to primitive variables (h, u, v) , trained by Adam at 10^{-3} for 20000 iterations with gradient clipping at 1.0, on 20000 collocation points resampled each step, with the initial condition imposed as a penalty at every cell centre weighted 10 against the residual. Following the source paper, the input and output layers stay dense and only the square hidden weights are factorised. The Fourier-feature baseline uses 32 frequencies. The finite-volume-informed entry predicts cell averages on the solver grid at 41 collocation times and is trained against the residual of the discrete SSP-RK2 update.

6.2 Results

Table 3 gives all five benchmarks, five seeds each. On the three shock-dominated cases no spectral regime separates from an unmodified network by more than about 10% in L^1 depth: on the step release every entry but the finite-volume-informed one falls between 1.03 and 1.13×10^3 , inside the unconstrained baseline’s own seed spread. The unconstrained network sits a factor of 4.1 to 4.4 above a classical scheme at the same resolution.

The two benchmarks that do separate separate against the method. On the Gaussian mound — the only smooth initial condition in the set, and the case closest to the regime the source paper validates in — exact orthogonality reaches $(9.69 \pm 1.8) \times 10^2$ against the unconstrained network’s

Table 3: L^1 depth error at the final output time on the 128^2 evaluation grid, mean $\pm$ s.d. over 5 seeds. Each column carries its own factor. Section 6.5 gives the reference uncertainty for the wet, step and Gaussian columns; it exceeds several of the differences here.

Entry	Circular, wet ($\times 10^3$)	Circular, dry ($\times 10^3$)	Step ($\times 10^3$)	Gaussian mound ($\times 10^2$)	Three humps ($\times 10^2$)
Classical, 128^2	0.391	0.380	0.248	0.064	0.797
Unconstrained	1.602 ± 0.012	1.681 ± 0.076	1.028 ± 0.11	3.83 ± 0.22	8.87 ± 0.68
Fourier features	1.681 ± 0.11	1.838 ± 0.12	1.124 ± 0.068	4.03 ± 0.48	7.96 ± 1.5
Penalised spectral	1.662 ± 0.12	1.602 ± 0.054	1.106 ± 0.047	4.69 ± 0.55	7.49 ± 1.1
Exactly orthogonal	1.725 ± 0.13	1.507 ± 0.14	1.054 ± 0.071	9.69 ± 1.8	9.19 ± 2.3
Spectrum only	2.347 ± 0.11	2.314 ± 0.19	1.126 ± 0.015	9.16 ± 0.34	6.08 ± 1.1
FV-informed	14.81 ± 2.4	9.32 ± 0.54	13.65 ± 2.7	44.0 ± 5.6	8.66 ± 0.22

$(3.83 \pm 0.22) \times 10^2$, and spectrum-only training $(9.16 \pm 0.34) \times 10^2$. That is a factor of 2.5 in the wrong direction, it is far outside the seed spread, and Section 6.5 puts the reference uncertainty on this benchmark at 3.8×10^2 , so the gap is resolvable where smaller ones are not. The penalised regime is intermediate at $(4.69 \pm 0.55) \times 10^2$. Constraining the factor hurts most where the target is smooth.

On the three-hump case with Manning friction the ordering inverts: spectrum-only training is the best neural entry at $(6.08 \pm 1.1) \times 10^2$ and exact orthogonality the worst at $(9.19 \pm 2.3) \times 10^2$. We do not read this as support for the spectral family, because every entry on that benchmark loses between 63% and 90% of its initial mass over the 20 s run — against 10^{-3} or below on the two-second cases — so all six models are far outside any regime where a depth error is meaningful. The case is reported for completeness and should not be used to rank methods.

The finite-volume-informed network collapses on the four short cases, matching the behaviour reported for it in [9].

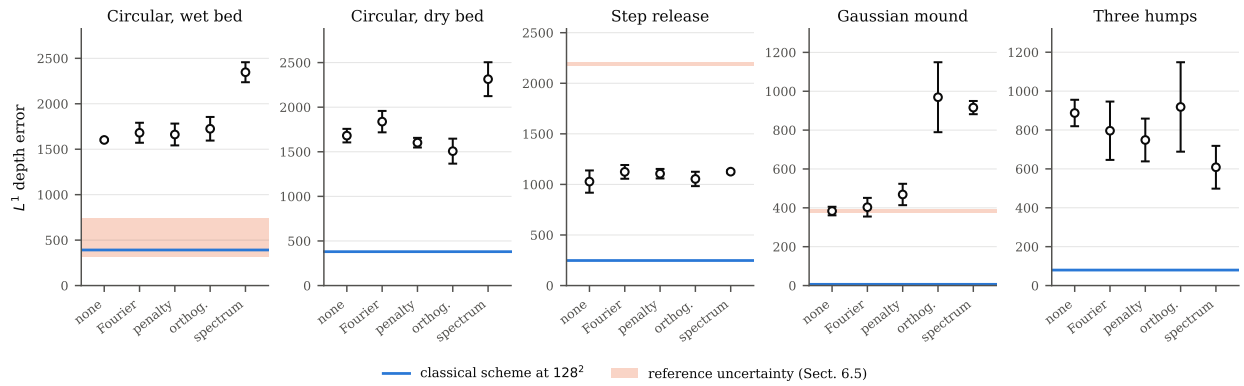


Figure 2: L^1 depth error by benchmark, mean $\pm$ s.d. over 5 seeds; the finite-volume-informed entry is off-scale and omitted. The blue line is a classical scheme at the evaluation resolution. The orange band, where present, spans the disagreement between our reference and two independently implemented schemes on the same initial condition (Section 6.5); on the step release it sits above every neural result, and on the Gaussian mound it is the width of the vanilla network’s own error. Only the Gaussian mound separates the regimes, and it separates them against the constrained ones.

6.3 Does the penalty weight matter here?

The first of the source paper’s stated limitations is that w_U must be tuned per equation, and no published value exists for shallow-water flow. We bracketed it over three decades on the two circular benchmarks with the corrected Eq. (1) penalty, five seeds per cell (Table 4).

Table 4: L^1 depth error against w_U over three decades, five seeds, using the 2-norm penalty of Eq. (1). The spread across the whole range is smaller than the spread across seeds within any one cell.

w_U	Circular, wet bed	Circular, dry bed
10^{-3}	$(1.578 \pm 0.025) \times 10^3$	$(1.765 \pm 0.032) \times 10^3$
$1/35000 = 2.9 \times 10^{-5}$	$(1.601 \pm 0.12) \times 10^3$	$(1.709 \pm 0.056) \times 10^3$
10^{-6}	$(1.549 \pm 0.069) \times 10^3$	$(1.643 \pm 0.050) \times 10^3$

Three orders of magnitude in w_U move the error by 3% on the wet bed and 7% on the dry bed, both inside the seed-to-seed spread. On these problems the tuning burden the source paper identifies does not exist, because the knob does not do anything. That is a negative answer to their first limitation in this regime rather than a solution to it: there is nothing to adapt. It also says the penalty is not the active ingredient here, which is consistent with the whole spectral family sitting within 10% of an unmodified network on the same benchmarks.

This sweep supersedes one reading we drew from the earlier, Frobenius-penalty run and now retract. There, the penalised network placed the radial front on the dry bed at 3.5 ± 4.1 m from the reference against the unconstrained network’s 21.2 ± 8.6 m, which we took as a local advantage invisible to the integrated error. Under the corrected penalty the same quantity is 14.8 ± 6.6 m at $w_U = 1/35000$ and between 14.8 and 17.0 m across the three decades, overlapping the unconstrained baseline within one standard deviation. The apparent front-position advantage belonged to the stronger Frobenius penalty, not to the published method, and it does not survive. Integrated depth error was almost unaffected by the same change (1.662×10^3 against 1.601×10^3 on the wet bed), so the norm choice matters for where the shock is placed and not for the depth field as a whole.

6.4 What the regimes cost

The second stated limitation is the penalty’s training overhead, which the source paper puts at 1.28 to 1.83 times vanilla across its width and collocation sweeps. Table 5 gives wall-clock training time over all 25 runs per entry on one Quadro RTX 8000.

Table 5: Mean training time per run, 25 runs per entry (5 benchmarks $\times$ 5 seeds), identical iteration budgets. The penalty is nearly free here; the penalty-free alternative is the expensive one.

Entry	Mean training time (s)	Relative to unconstrained
Unconstrained	861	1.00 $\times$
Spectrum only	862	1.00 $\times$
Penalised spectral	884	1.03 $\times$
Fourier features	946	1.10 $\times$
Exactly orthogonal	1226	1.42 $\times$
FV-informed	2048	2.38 $\times$

The overhead the source paper reports does not reproduce in this setting: the penalty costs 3%, not 28–83%. One 2-norm of a 128×128 matrix per factorised layer per step is negligible against 20000 collocation points, and we suspect the published figures reflect a smaller problem where the penalty is a larger share of the step, or a different implementation of the norm.

The direction of the comparison matters more than the absolute numbers. Removing the penalty by constraining the factor exactly costs 42%, because the matrix-exponential retraction runs on every forward pass. So on cost grounds the penalty-free route is worse than the penalty, not better, and the motivation for eliminating w_U has to rest on the tuning burden alone — which Section 6.3 shows does not exist on these problems either. Spectrum-only training is free relative to the baseline, as expected from training n parameters per layer instead of n^2 , but it buys nothing in accuracy.

6.5 How far the reference can be trusted

Every number in Table 3 is measured against one solver’s output. To find out what that is worth, we compare the reference against an independent implementation: the HLL, Lax–Wendroff and MUSCL–Rusanov runs produced for [9] on a 501^2 node grid for the same initial conditions, interpolated onto the evaluation grid. Those runs store depth only, so the comparison is on depth.

Table 6: L^1 depth disagreement between our 512^2 HLLC reference and three independently implemented schemes on the same initial condition, against our own discretisation error and the unconstrained network’s error. Read down a column: where the middle entries exceed the last, the benchmark cannot resolve differences of the size Table 3 reports.

	Circular, wet bed	Gaussian mound	Step release
Our classical at 128^2	3.914×10^2	6.357×10^0	2.476×10^2
vs. their HLL	3.111×10^2	3.763×10^2	2.169×10^3
vs. their MUSCL–Rusanov	7.395×10^2	3.949×10^2	2.208×10^3
vs. their Lax–Wendroff	7.551×10^3	3.435×10^3	9.349×10^3
Unconstrained network	1.602×10^3	$\sim 3.9 \times 10^2$	1.028×10^3

Three regimes appear. On the wet-bed column the reference is vindicated: it agrees with an independent HLL implementation to 3.11×10^2 , tighter than its own 128^2 discretisation error of 3.91×10^2 , and both are far below the 1.60×10^3 network error. Differences between network variants on that benchmark are real. On the Gaussian mound the two solver families disagree by 3.76×10^2 and the network error is 3.9×10^2 — the same number — while our own discretisation error is sixty times smaller at 6.36. On the step release the solver families differ by 2.2×10^3 , twice the network error. No ranking of network variants on those two benchmarks survives a change of reference solver.

Lax–Wendroff is the outlier throughout, which is expected of Lax–Wendroff with artificial viscosity at a shock and is a statement about that scheme rather than about ours. Nothing here says our reference is wrong; it is the most refined and least diffusive of the four, and on the wet-bed column it is confirmed. What it says is that a comparison table of this kind has a resolution floor that varies by benchmark, and that the floor has to be quoted next to any margin claimed from it. We have not seen this check performed in the PINN-for-conservation-laws literature, where a single fine-grid run is the usual reference.

7 Limitations

Five seeds per cell resolve a factor of two, not the 10% differences that make up most of Table 3; only the Gaussian-mound column separates by more than the spread. Section 6.5 bounds what the comparison can resolve at all and should be read before Table 3, not after it. The three-hump case is reported but not interpretable: every model there loses most of its mass.

The factor-geometry measurement of Table 2 uses 3 seeds on one one-dimensional problem, one architecture and one optimiser, and it is the basis for the paper’s central claim about what the penalty does. It should be repeated on the shallow-water networks, where the per-run factor diagnostics are now recorded but not yet analysed.

The singular-value box of Section 5.3, which is what the refuted prediction points to, is not implemented. Until it is, the claim that the penalty acts by pulling the whole spectrum toward 1 rests on one measurement of D and one failed alternative, both on the advection case.

Two of the five contributions in the project proposal have no section here. The pointwise condition-number field, which is the diagnostic that would explain *where* on the shock the spectral variants fail rather than only that they do, is not built. Neither is the Navier–Stokes leg — the Taylor–Green case the source paper reports and the lid-driven cavity — which is what would connect these results back to the setting [8] validates in. Cost and memory were recorded per run but are not tabulated, so the second stated limitation of the source paper is not yet answered quantitatively.

8 Conclusion

The orthogonality penalty of [8] is load-bearing on their own advection problem and is not doing what the paper says it does. At the published weight the trained factor is rank-deficient rather than orthogonal, and the trained singular values are not the effective weight’s spectrum, yet removing the penalty destroys the accuracy gain and enforcing exact orthogonality recovers only about half of it. Replacing the penalty with a bound on the factor’s largest singular value — the weaker constraint that reading suggests — is the worst of the six regimes we tried. That rules the scale-bound reading out rather than replacing it: the bounded networks match the penalised ones on the orthogonality defect D while missing the effective spectrum by two further orders of magnitude, so D is too coarse a statistic to characterise what the penalty does, and what it does instead remains open.

Moving to shock-dominated shallow-water flow, outside the reach of the source paper’s Theorem 2.1, the picture is flatter and in one place reversed. On the three shock cases no spectral regime separates from an unmodified network by more than about 10% in integrated depth error, and the penalty weight can be moved three decades without measurable effect, so the tuning problem the source paper poses does not arise. On the one smooth case in the set, exact orthogonality and spectrum-only training are a factor of 2.5 *worse* than an unmodified network, by a margin that exceeds the reference uncertainty. Whatever the reparameterisation contributes on smooth one-dimensional targets does not transfer, and constraining the factor hurts most where the target is smoothest.

Establishing any of this required checking the reference against an independent solver family, which on two of three benchmarks disagrees with it by more than the differences under test. We would encourage that check as routine practice: a single fine-grid run is the usual reference in this literature, and its uncertainty is rarely quoted.

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